# Experiment - 6

## Develop a simple calculator using XMLRPC

**Aim:**

To develop a simple calculator application using XMLRPC in python implements a client server architecture

**Procedure:**

i) create server script 'calc_server.py' using simple XMLRPC server

ii) Define architecture function using server register function

iii) register all function using server register function

iv) Enable introspection with server.regis

v) start the server on port 8000

vi) create client script 'calc_client.py' using XMLPRC client server proxy

vii) connect to the server URL 'http://localhost:8000'

command / code:

```
# ... calc_server.py ...

from xmlrpc.server import SimpleXMLRPCServer

def add(a,b): return a+b

def sub(a,b): return a-b

def mul(a,b): return a*b

def div(a,b):
    if b==0: raise ValueError
    return a/b

def pow(a,b): return a**b

def modulo(a,b): return a%b

server = SimpleXMLRPCServer(('localhost', 8000),
                            logRequests=False)

for fn in [add, sub, mul, div, pow, modulo]:
    server.register_function(fn)

print("Calculation server running on
      port 8000")
```

xiii) call remote functions through the proxy object

ix) Display the result from the server

x) Test all arithmetic operator

Result: 8/10 13/X

The XMLRPC calculator application was successfully developed. The server exposed arithmetic operation and the client called them remotely. All mathematical operation were verified correctly